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CO1-Assignment Tool 1
Application based questions
Course code: DSA0308
Course Name: Natural Language Processing

Reg No: 192325087 **Name:** G.Yaswitha

1. Regular Expressions - Resume Information Extraction

- Start the program.
- Import the regular expression module using import re.
- Store the resume details in text format or user input.
- Extract the candidates name using r"Name:\s*(.*)".
- Extract the email address using r"[\w.-]+@[\w.-]+\.\w+".
- Extract the mobile number using r"\b[6-9]\d{9}\b".
- Detect technical skills using r"Python|Java|SQL|Machine Learning|NLP".
- Extract years of experience using r"Experience:\s*(\d+)".
- Display the candidates name, email, mobile number, skills, and experience.
- Check whether the candidate has atleast 2 years of experience and python skills.
- Display the Eligible Status as Eligible or Not Eligible.
- Stop the program.

Expected Output:
Name: Arun kumar
Email: arun@gmail.com
Mobile: 9876543210
Skills: ['Python', 'Java', 'SQL']
Experience: 3 years
Status: Eligible

2. Regular Experssions- Product Search System

- Start the program.
- Import the regular expression module using import re.
- Store the product names in a list.
- Get the search keyword from the user.
- Perform an extract keyword search using re.fullmatch(re.escape(keyword), product, re.I).
- Perform a prefix search using re.match(re.escape(keyword), product, re.I).
- Perform a suffix search using re.search(re.escape(keyword) + r"$", product, re.I).
- Perform a partial keyword search using re.search(re.escape(keyword), product, re.I).
- Use re.I to perform case- insensitive searches.
- Display all matching product names for each search type.
- Count and display the total number of matching products for each search
- Stop the program.

Sample Input:
Enter search keyword: Python

Expected Output:
Exact Search:
Matching products: []
Total matches: 0

Prefix Search:
Matching products: ['Python Book', 'Python Course']

Total matches: 2

Suffix Search:
Matching products: []
Total matches: 0

Partial Search:
Matching products: ['Python Book', 'Python Course']
Total matches: 2

3. Regular Expressions - University Registration System

- Start the program.
- Import the regular expression module using import re.
- Get the students register number, institutional email, course code, semester, mobile number.
- Validate the register number using r"^[A-Z]{2,3}\d{2}[A-Z]?\d{3,4}$".
- Validate the institutional email using using r"^[\w.-]+@saveetha\.com$".
- Validate the course code using r"^[A-Z]{2,4}\d{3}$".
- Validate the semester using r"^[1-8]$".
- Validate the mobile number using r"^[6-9]\d{9}$".
- Use re.fullmatch() to check whether each input matches its required pattern.
- Display a valid or invalid message for each field.
- If all fields are valid, display registration successful; Otherwise, display registration failed.
- Stop the program.

Sample Input:
Register number: 192325087B
Institutional email: yaswithag@saveetha.com
Course code: DSA0308
Semester (1-8): 7
Mobile number: 9876543210

Expected Output:
Register Number: Valid
Email: Valid
Course Code: Valid
Semester: Valid
Mobile: Valid
Registration Status: Successful